Internet Programming

22/10/2026

Name : Harshad Sanjay Teli.
Roll No : 15
Batch : B1

Function and Output Methods

Assignment-9(Function and Output Methods)

1. Create a Calculator Using Functions

Write separate functions for:

- Addition
- Subtraction
- Multiplication
- Division

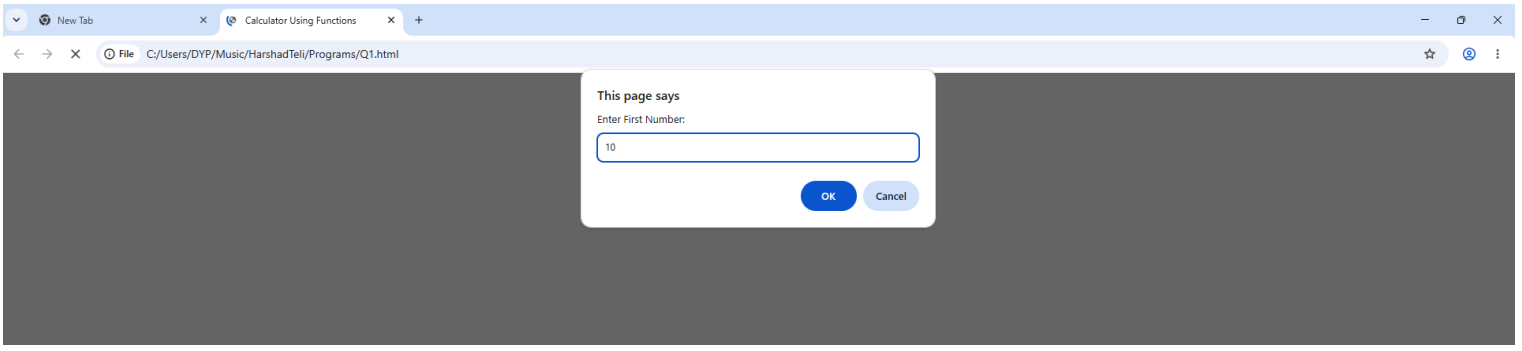
Then call them based on user choice.

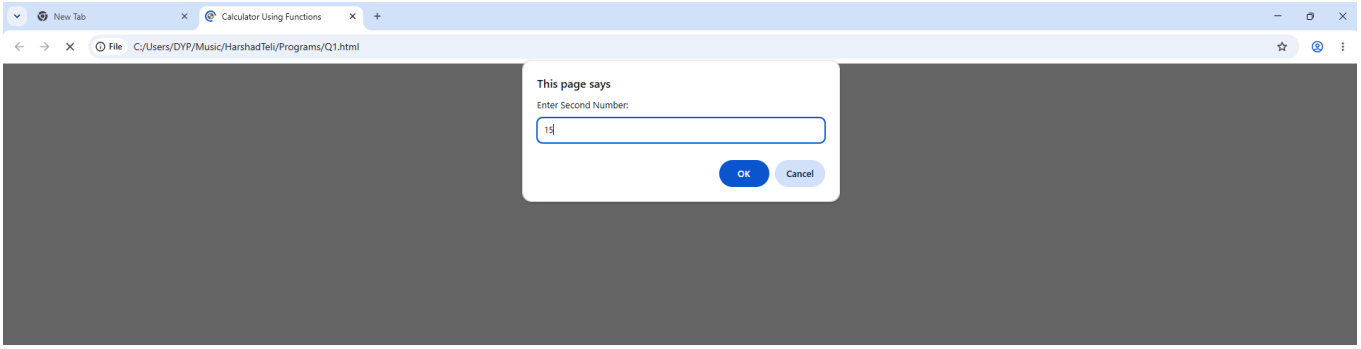
Input:

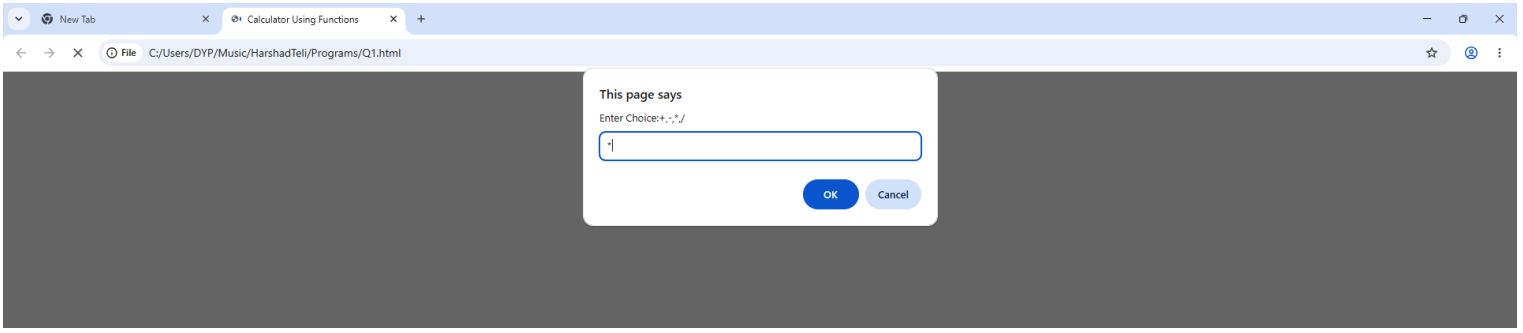
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Calculator Using Functions</title>
</head>
<body>
    <script type="text/javascript">
    function addition(a,b){
        return a+b;
    }
    function subtraction(a,b){
        return a-b;
    }
    function multiply(a,b){
        return a*b;
    }
    function divide(a,b){
```

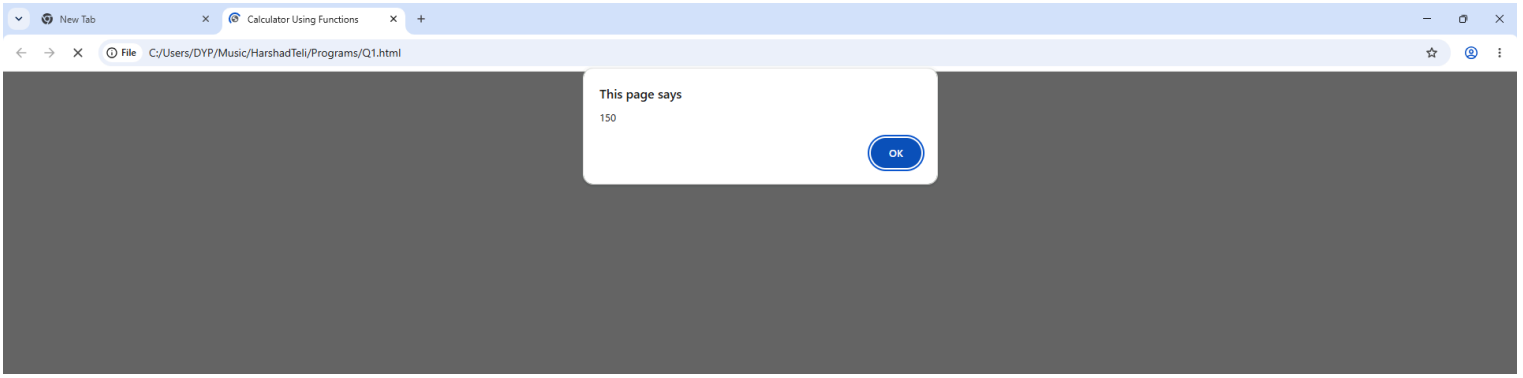
```
        return a/b;
    }
    let a = Number(prompt("Enter First Number:"));
    let b = Number(prompt("Enter Second Number:"));
    let choice = prompt("Enter Choice:+,-,*,/");
    if(choice=='+'){
        alert(addition(a,b));
    }else if(choice=='-'){
        alert(substraction(a,b));
    }else if(choice=='*'){
        alert(multiply(a,b));
    }else if(choice=='/'){
        alert(divide(a,b))
    }
    else{
        alert("Invalid Choice")
    }
    </script>
</body>
</html>
```

Output:









Rewrite the same program using **arrow functions**.

Input:

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Calculator using Arrow functions</title>
</head>

<body>
    <script type="text/javascript">
        let add = (a, b) => a + b;
        let sub= (a, b) => a - b;
        let mul = (a, b) => a * b;
        let div = (a, b) => a / b;
         let a = Number(prompt("Enter First Number:"));
    let b = Number(prompt("Enter Second Number:"));
    let choice = prompt("Enter Choice:+,-,*,/");

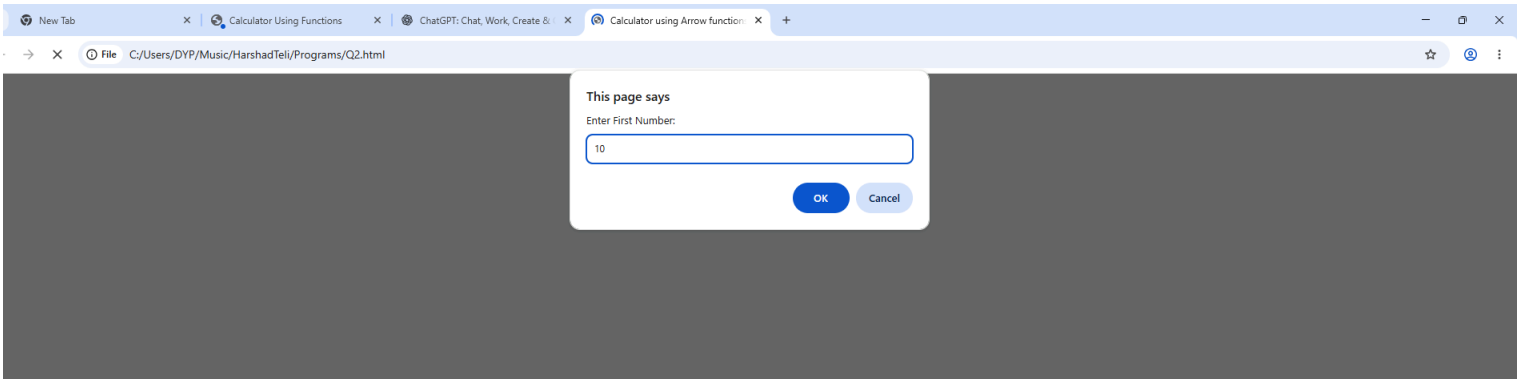
    if(choice=='+'){
        alert(add(a,b));
    }else if(choice=='-'){
        alert(sub(a,b));
```

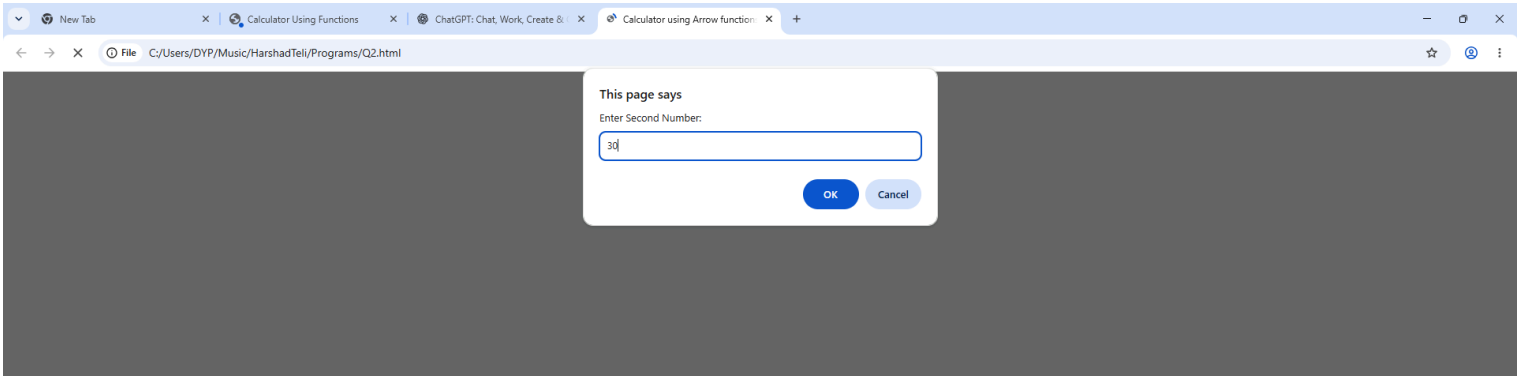
```
    }else if(choice=='*'){
        alert(mul(a,b));
    }else if(choice=='/'){
        alert(div(a,b));
    }
    else{
        alert("Invalid Choice");
    }

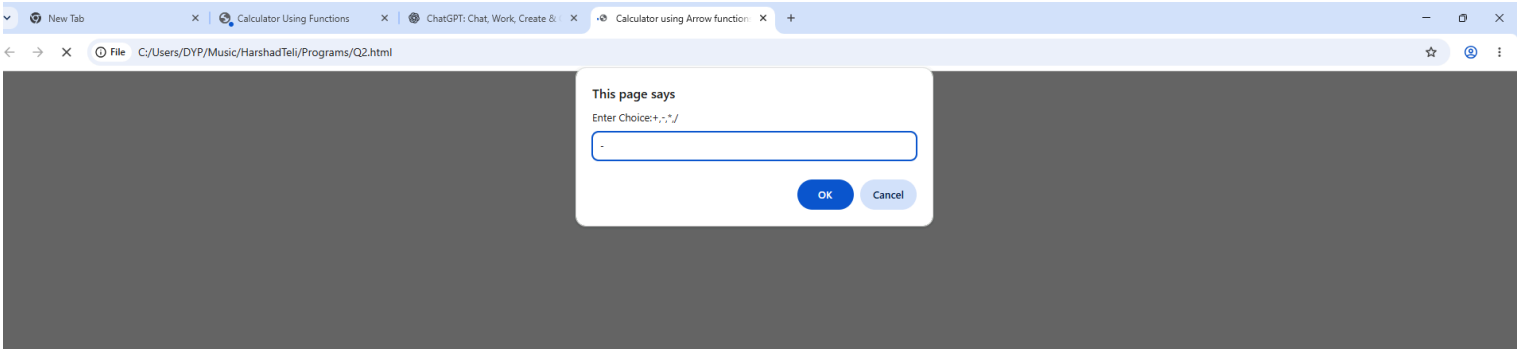

    </script>
</body>

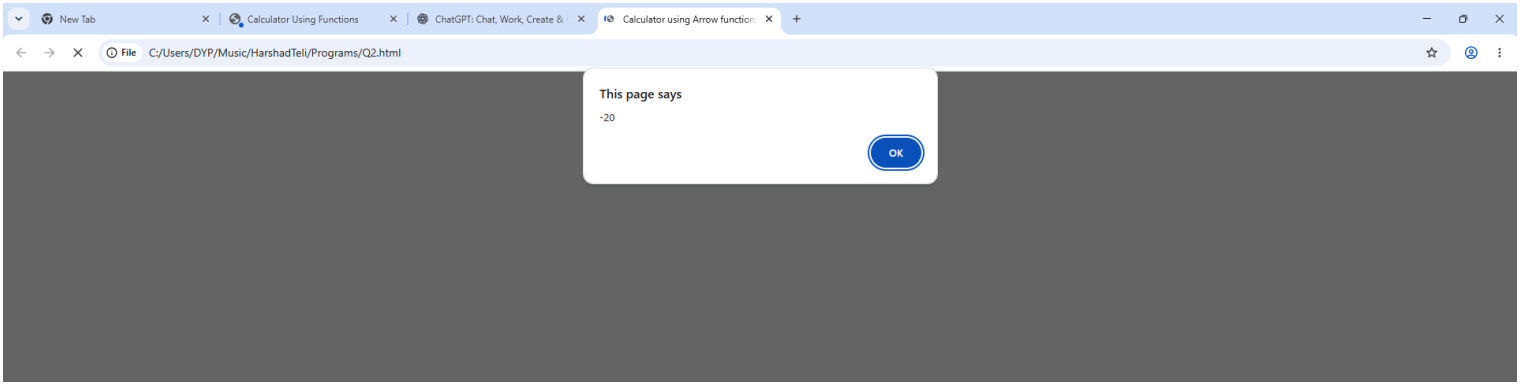
</html>
```

Output:








New Tab
Calculator Using Functions
ChatGPT: Chat, Work, Create &
Calculator using Arrow function
File
C:/Users/DYP/Music/HarshadTeli/Programs/Q2.html
This page says
-20
OK

2. **Array Processing.**

Take an array of numbers and write:

- A function to return all even numbers
- A function to return sum of all numbers
- A function to return maximum number something more on array Count How Many Numbers Are Positive, Negative, and Zero.

Input:

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Array Processing</title>
    <style>
        body {
            background-color: black;
            color: whitesmoke;
        }
    </style>
</head>

<body>

    <script type="text/javascript">
        let p = 0; n = 0, z = 0;
        let arr = [10, -5, 20, 0, -8, 15, 0];
        function even() {
            let result = [];
            for (let i = 0; i < arr.length; i++) {
                if (arr[i] % 2 == 0) {
                    result.push(arr[i]);
                }
            }
            return result;
        }
```

```
        function sum() {
            let s = 0;
            for (let i = 0; i < arr.length; i++) {
                s = s + arr[i];
            }
            return s;
        }
        function max() {
            let max = arr[0];
            for (let i = 0; i < arr.length; i++) {
                if (arr[i] > max) {
                    max = arr[i];
                }
            }
            return max;
        }
        function count() {

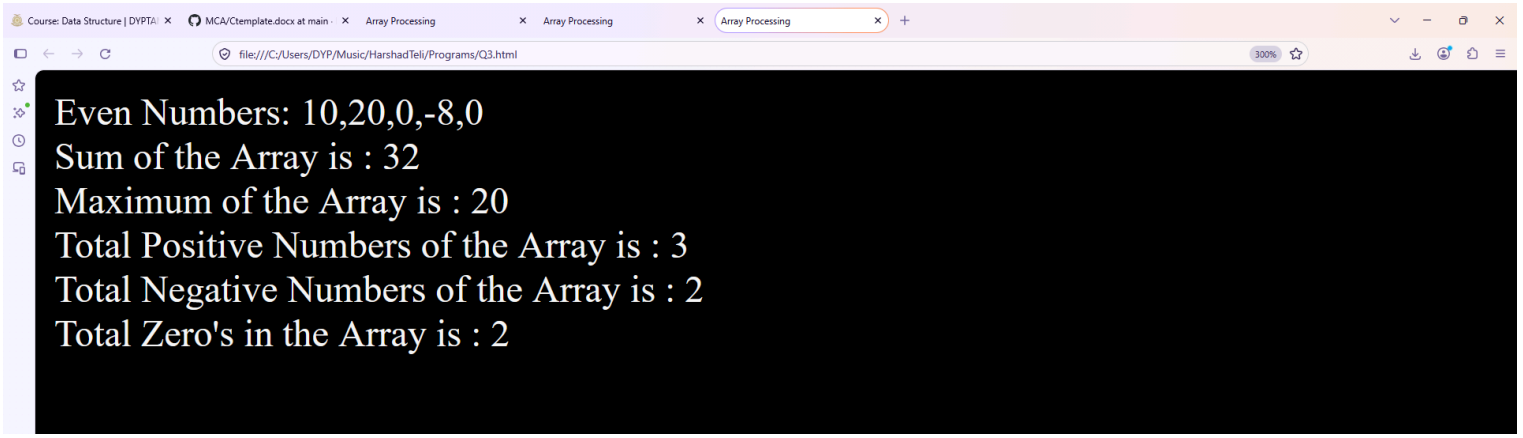
            for (let i = 0; i < arr.length; i++) {
                if (arr[i] > 0) {
                    p++;
                }
                else if (arr[i] < 0) {
                    n++;
                }
                else {
                    z++;
                }
            }

        }
        count();
        document.write("Even Numbers: " + even() +
"<br>");
```

```
        document.write("Sum of the Array is : " + sum() +
"<br>");
        document.write("Maximum of the Array is : " +
max() + "<br>");
        document.write("Total Positive Numbers of the
Array is : " + p + "<br>");
        document.write("Total Negative Numbers of the
Array is : " + n + "<br>");
        document.write("Total Zero's in the Array is : "
+ z + "<br>");
    </script>
</body>

</html>
```

Output:



3. Student Information Form (without HTML form tag)

Using only:

- prompt()
- innerHTML
- alert()

Ask the user:

- Name
- Age
- Course
- College

Then display the details in a in **table format** using innerHTML.
If age < 18 → show alert **"Not eligible"**.

Input:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Information Form</title>
</head>
<body>
    <div id="result"></div>
        <script type="text/javascript">
            let  name= prompt("Enter Your Name:");
            let age=prompt("Enter Age:");
            let course=prompt("Enter Course:");
            let college=prompt("Enter College:");

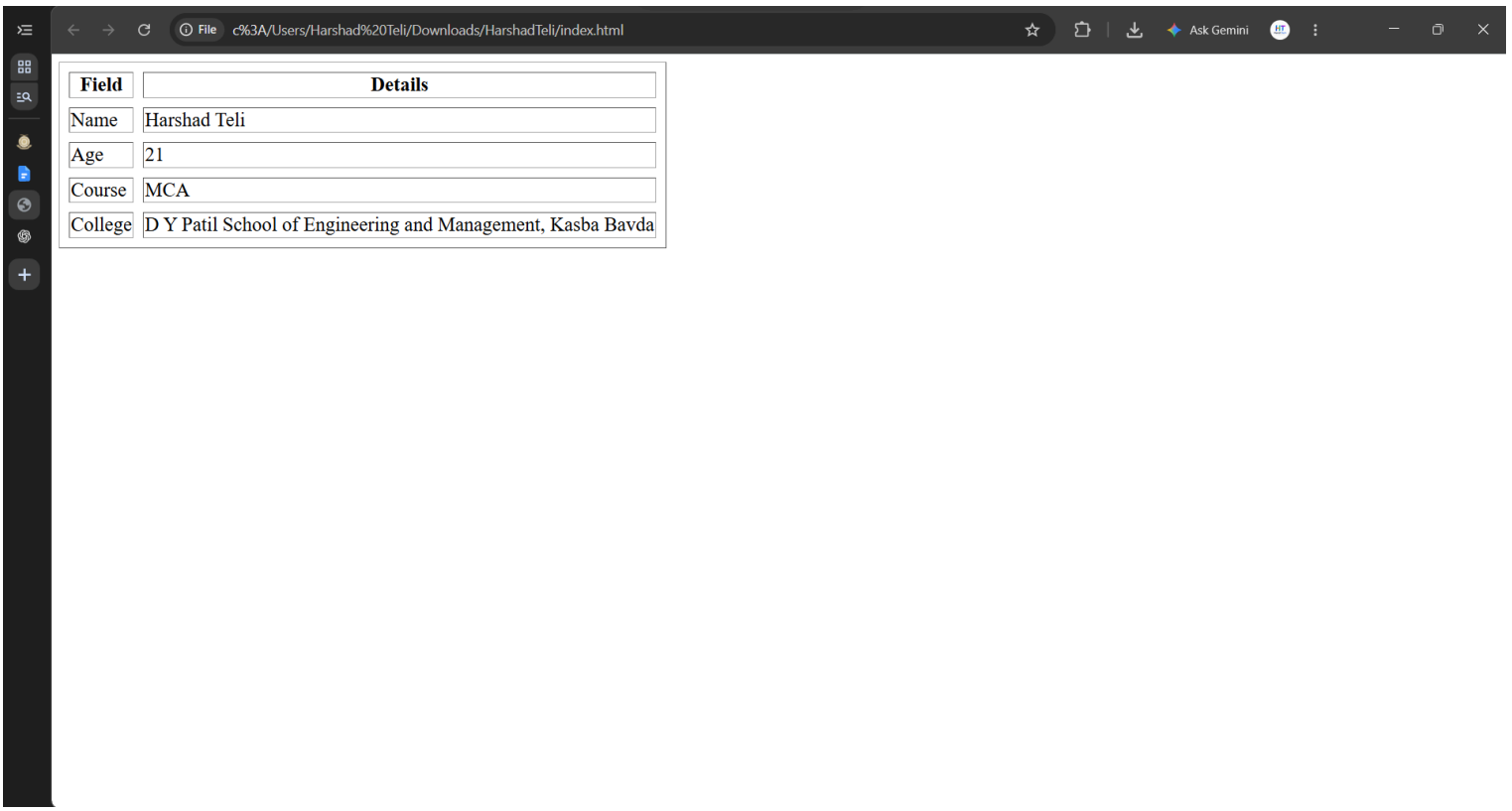
            if(age<18){
                alert('Not Eligible');
            }
            else{
                document.getElementById("result").innerHTML=`
                 <table border='1' cellspacing='10'>
            <tr>
                <th>Field</th>
                <th>Details</th>
            </tr>
```

```
            <tr>
                <td>Name</td>
                <td>${name}</td>
            </tr>
            <tr>
                <td>Age</td>
                <td>${age}</td>
            </tr>
            <tr>
                <td>Course</td>
                <td>${course}</td>
            </tr>
            <tr>
                <td>College</td>
                <td>${college}</td>
            </tr>
        </table>`;

            }
        </script>

</body>
</html>
```

Output:



Field	Details
Name	Harshad Teli
Age	21
Course	MCA
College	D Y Patil School of Engineering and Management, Kasba Bavda

4. Login Authentication

Create a program:

- Ask user for **username** and **password** using prompt().
- If username = "admin" AND password = "1234" → display "Login Successful" using innerHTML.
- Else → show an **alert("Invalid Login!")**.

Use confirm("Do you want to try again?") to repeat the login attempt.

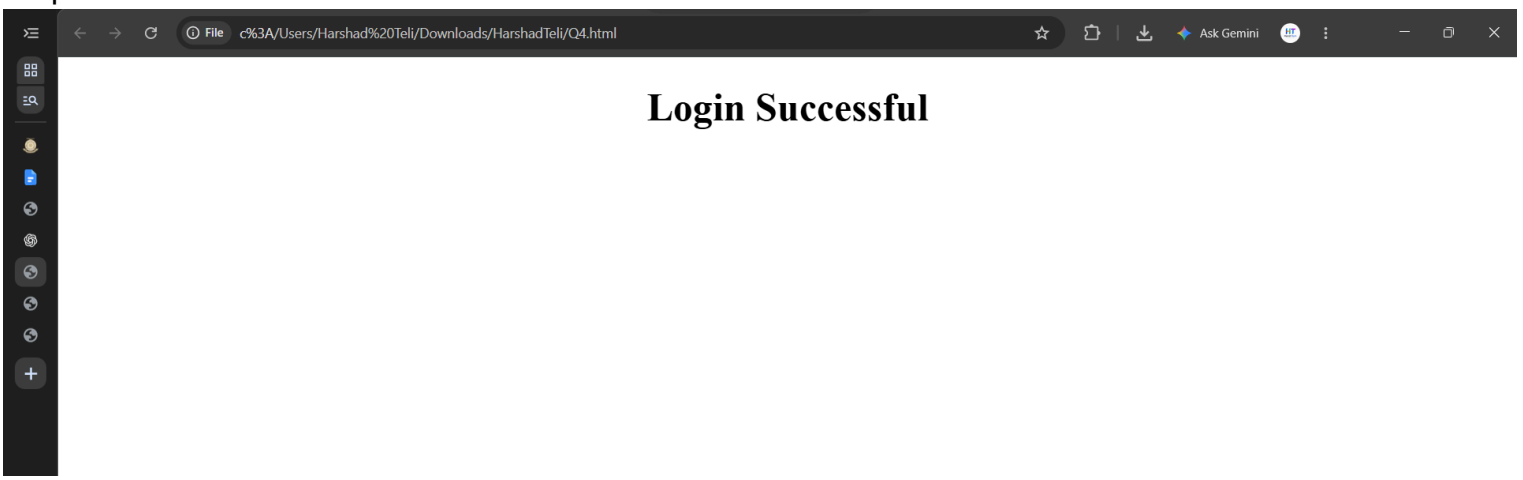
Input:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Login Aunthentication</title>
</head>
<body>
    <div id="result"></div>
    <script type="text/javascript">
        let username,password,again;
```

```
        do{
            username=prompt("Enter Username:");
            password=prompt("Enter Password:");
            if(username==='admin' && password==='admin1234'){
                document.getElementById("result").innerHTML=`
                <center>
                    <h1>Login Successful</h1>
                </center>`;
            }
            else{
                alert("Invalid Login");
                again=confirm("Do you want to try again?");
            }
        }while(again);
    </script>

</body>
</html>
```

Output:



5. Confirm Navigation

Create a “Go to Google” button.
On clicking:

- Show a confirm("Do you really want to leave this page?")
- If user clicks OK → redirect to Google.
- If Cancel → show **innerHTML = "Stayed on the page"`**

Input:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Confirm Navigation</title>

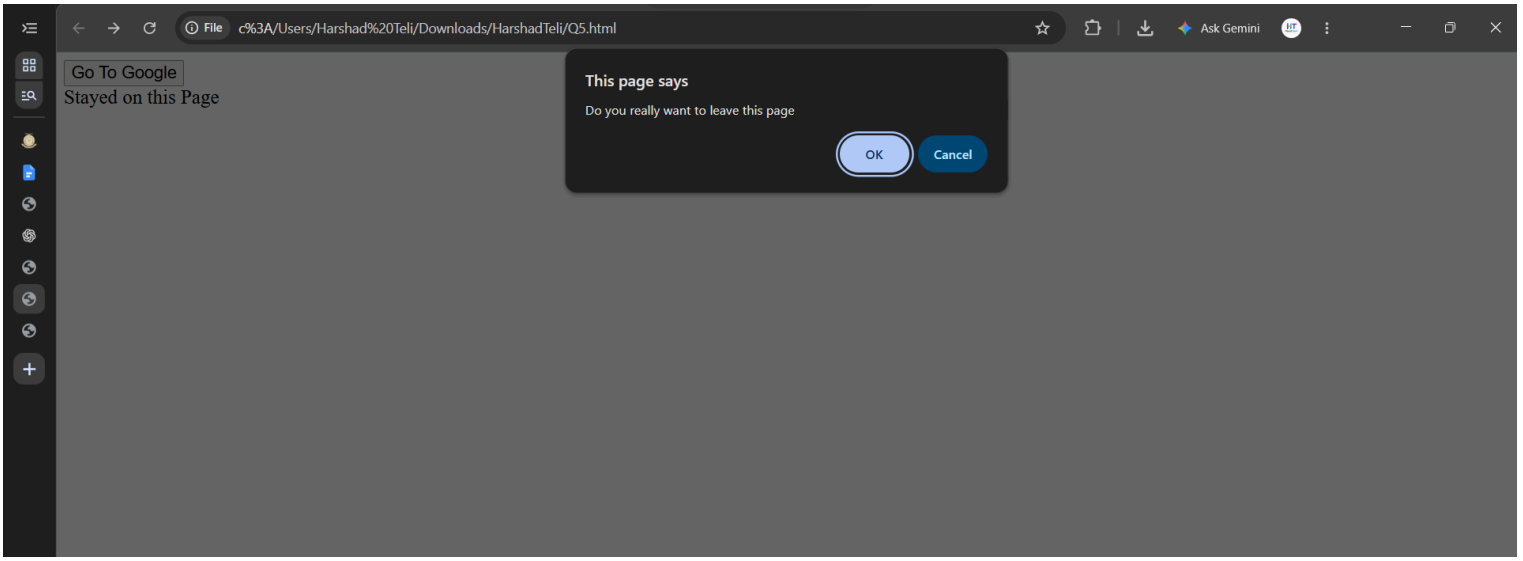
</head>

<body>
```

```
    <button onclick="goToGoogle()">Go To Google</button><br>
    <div id="result"></div>
    <script type="text/javascript">
        function goToGoogle(){
            let choice=confirm("Do you really want to leave
this page");
        if(choice){
            window.location.href="https://www.gooogle.com";
        }else{

document.getElementById("result").innerText="Stayed on this
Page";
        }
        }
    </script>
</body>
</html>
```

Output:



6. Dynamic Background Color Chooser

Ask user via prompt():
"Enter background color (red/blue/yellow/green):"
If color is valid → apply it to <body> using style + innerHTML message.
If invalid → show alert "Color not available".

Input:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Dynamic Color Chooser</title>
</head>
<body>
    <div id="result"></div>
    <script type="text/javascript">
```

```
        let color=prompt('Enter
Color:'+'\n(Red,Blue,Yellow,Green)');

if(color=="red"||color=="blue"||color=="yellow"||color=="green")
{
            document.body.style.backgroundColor=color;

document.getElementById("result").innerText="Background Color is
Changed To "+color;

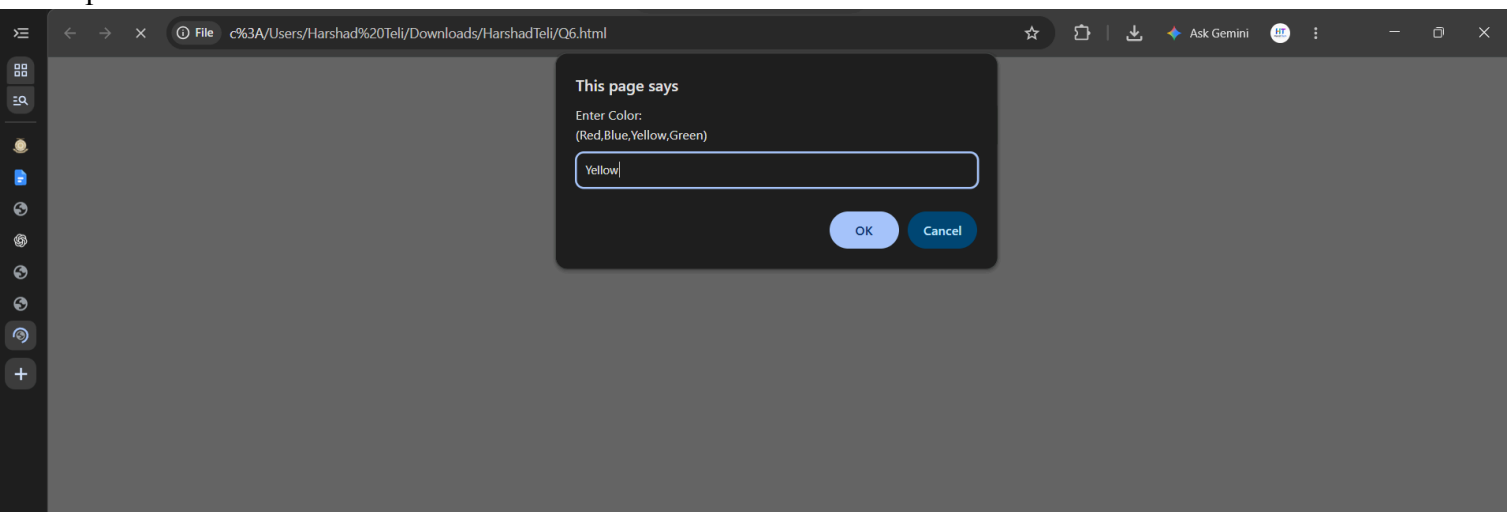
        }else{
            alert("Color not available");

        }

    </script>

</body>
</html>
```

Output:




File
c%3A/Users/Harshad%20Teli/Downloads/HarshadTeli/Q6.html
Ask Gemini
Background Color is Changed To yellow